SPUDNAUT: The Great Starch Migration

Act I: The Great Tuber Crisis (Welcome & The Spud Exam)

- **Visuals (Welcome Slide):** A sprawling, sepia-toned planet Earth completely covered in mountains of potatoes, squinting under a tired sun.
- **Intro Text / Lore:**
  "Year 3042. Earth's soil has given its last harvest. Overfarming has spawned an unprecedented crisis: we are drowning in potatoes. To save our species from running out of space, the Spudnaut Space Agency has initiated a desperate mission. Ten elite tubers will be sent to the Moon or Mars to test planetary habitability. Will you be a hero, or will you end up as instant mashed potatoes?"
- **The Spud Exam (Assessment):**
  - *Gameplay:* Multiple-choice quiz covering space basics, atmospheric pressure, and tool handling (e.g., *"What happens to an unshielded potato in a cosmic vacuum? A) Instant french fry. B) Infinite sprout mode. C) Cryo-freeze."*).
  - **Branching Paths:**
    - **If Pass:** A triumphant animation plays of a potato in a tiny spacesuit walking across a stage receiving a diploma, stamped with a golden potato seal → *Proceed to Space Station.*
    - **If Fail:** A buzzing buzzer sounds. A drill sergeant potato shakes its head. The player enters the **"Remedial Spud Academy"** (quick mini-lessons on failed topics) → *Retake Exam until passed.*

Act II: Ground Control to Major Tuber (Orientation & Roles)

- **The Station Briefing:** The player arrives at the orbital hub, greeted by **Commander Russet**, an old, grizzled potato with a patch over one eye (made of a dried pea).
- **The Guide:** A holographic assistant named **S.P.U.D.I.E.** breaks down the 30-day trial rules, survival metrics (Oxygen, Starch Integrity, Morale), and tools.
- **Role Selection:** The player picks their professional spud specialty, which alters their missions and skill perks:
  1. **Farmer:** Expert in growing synthetic soil and managing water recycling.
  2. **Researcher:** Deals with pressure leaks and chemical toxicity tests.

3. **Traveller:** Handles surface exploration and deep-crater tasks.
4. Kulang pani

Act III: Houston, We Have a Starch-Emergency (The Launch & In-Flight Crisis)

- **The Flight Cinematic:** A gorgeous slide/animation sequence of the rocket blasting off, shaking violently, and breaking through the atmosphere into the starry void.
- **The In-Flight Emergency:** Warning sirens blare! Cosmic radiation has fried the primary navigation relay, or a micrometeoroid has punctured the potato-oil cooling line.
- **First Task (Among Us Style):** The player must rush to the engine room, navigate corridors, and complete a quick minigame (e.g., reconnecting wires or patching a hull breach with space-duct tape) to stabilize the ship before reaching the destination.

Act IV: Choosing Your Destiny & The 30-Day Surface Trial (Moon vs. Mars)

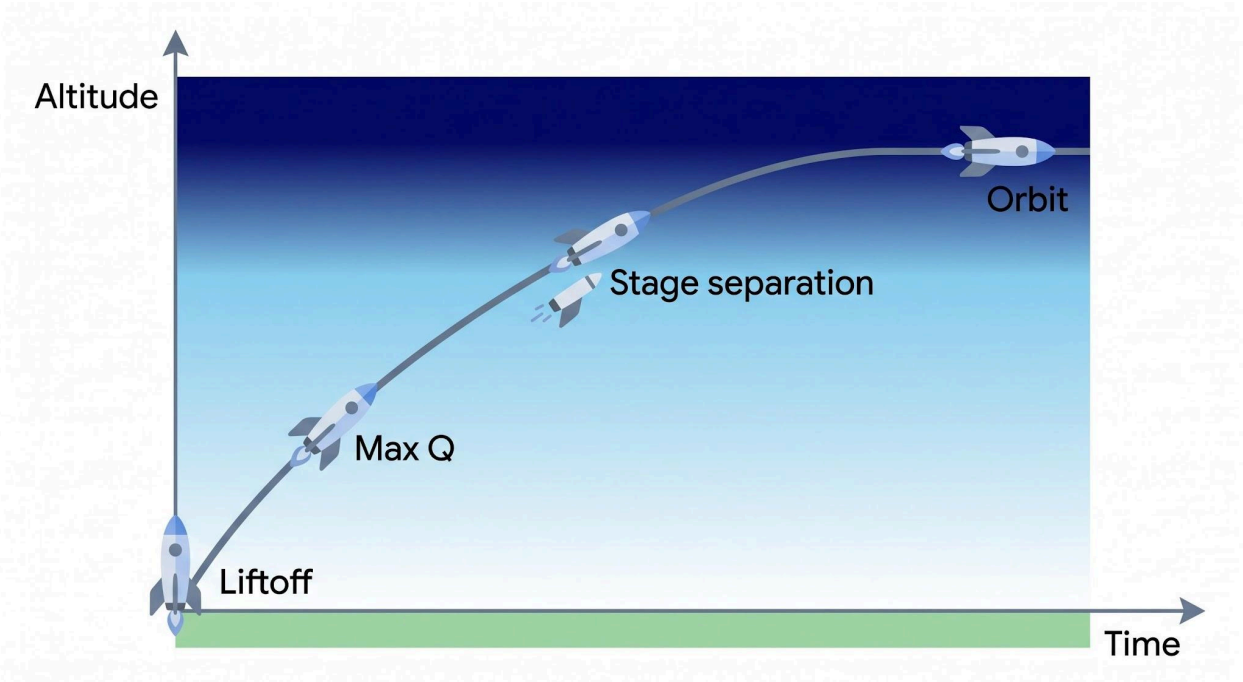
- **Destination Selection:** The player chooses whether to land on the **Moon** (low gravity, dusty, extreme temperature swings) or **Mars** (thin rust atmosphere, solar dust storms, longer days).
- **The 30-Day Cycle (Core Gameplay Loop):**
    - A calendar tracks Day 1 to Day 30.
    - Every few days features a mix of **turn-based decisions, card selection** (e.g., playing a "Thermal Blanket" card during a solar flare), and **Among Us-style tasks** (calibrating solar panels, testing soil ph, fixing oxygen scrubbers).
    - **Space Barriers & Factors:** Random events test the potato's resilience (e.g., space radiation causing "sprouting anxiety," equipment freezing, or loneliness requiring a morale boost).

Act V: The Verdict & Endings

At the end of Day 30, the station AI evaluates the player’s survival metrics based on their tasks and decisions:

- **Ending A: Mission Success (Passed)**
    - *Condition:* Met all survival and research requirements.

    - *Cutscene:* The pod launches back to Earth. The player lands to a massive ticker-tape parade of cheering potatoes holding signs. The hero potato is crowned with a golden foil helmet.
    - *Result:* Credits roll with an upbeat, triumphant chiptune/orchestral track.
- **Ending B: The Penalty Round (Failed / Extension)**
    - *Condition:* Fell short of requirements during the 30 days.
    - *Cutscene:* Commander Russet sighs over the radio. *"Your starch integrity is dropping, Spudnaut. You haven't proven habitability yet."*
    - *Gameplay:* The player is handed a **15-day penalty extension** in the deep-freeze crater sector of the planet, requiring high-stakes tasks to redeem themselves before a final evaluation.



- **1.Ignition & Liftoff:**Launchpad clear.

  The main engines ignite to generate thrust exceeding the rocket's total weight (a thrust-to-weight ratio greater than 1). The holding clamps release, and the rocket clears the launch tower vertically.

- **2.Pitch Maneuver & Gravity Turn:**Trajectory transition.

  Shortly after clearing the tower, the rocket tilts slightly (pitches) downrange. Earth's gravity naturally pulls the flight vector downward, turning vertical thrust into horizontal velocity while minimizing steering fuel consumption.

- **3.Max-Q (Maximum Dynamic Pressure):**Peak structural stress.

  As the vehicle accelerates through the lower atmosphere, dynamic pressure reaches its peak (a combination of high speed and air density). Engines briefly throttle down to reduce aerodynamic strain on the airframe.

- **4.Main Engine Cut-Off (MECO):**First-stage depletion.

  Once the primary booster exhausts most of its propellant, the first-stage engines shut down. The rocket is now near or above the denser layers of the atmosphere.

- **5.Stage Separation:**Mass reduction.

  Pneumatic pushers or explosive bolts detach the spent first stage. Shedding the heavy empty tanks and engines reduces overall mass, allowing the next stage to accelerate much more efficiently.

- **6.Second-Stage Ignition (SECO-1):**Upper-stage burn.

  The upper-stage vacuum engine ignites in space. Because atmospheric pressure is virtually zero, this stage uses a larger nozzle designed for maximum efficiency in a vacuum.

- **7.Payload Fairing Separation:**Exposing the spacecraft.

  Once the rocket ascends past the dense atmosphere (usually above 100 km), the protective nose cone (fairing) splits in half and jettisons, lightening the remaining vehicle.

- **8.Orbital Insertion:**Final speed reached.

  The upper stage burns until the payload achieves horizontal orbital velocity (around 28,000 km/h for Low Earth Orbit). Once reached, Second Stage Engine Cut-Off (SECO) occurs, and the satellite or spacecraft separates into its designated orbit.

The Space Launch System Explained | America's New Moon Rocket